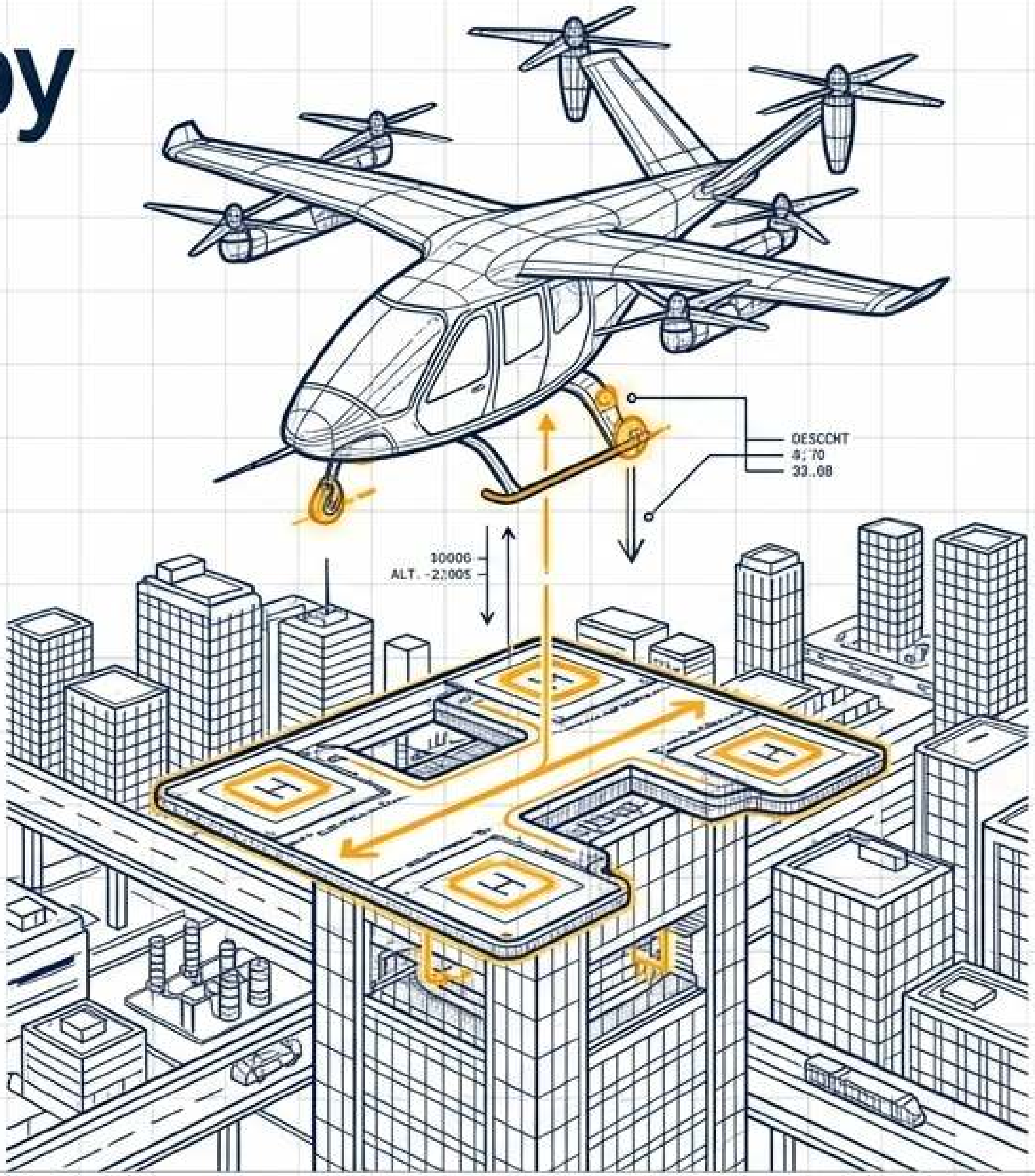
Engineering the Vertical Canopy

7.1 Urban Air Mobility & Vertiport Architecture

MASTERCLASS REFERENCE DOCUMENT

Distilling the physical, digital, and regulatory engineering required to scale 7.1 Urban Air Mobility (UAM) and Vertiport operations.



01 / EXECUTIVE OVERVIEW

02 / CORE TECHNICAL PRINCIPLES

03 / HARDWARE & SOFTWARE INTEGRATION

04 / REAL-WORLD APPLICATION

05 / FUTURE TRENDS

DEFINING THE URBAN AIR MOBILITY PARADIGM

SIMPLIFIED VEHICLE OPERATIONS (SVO)

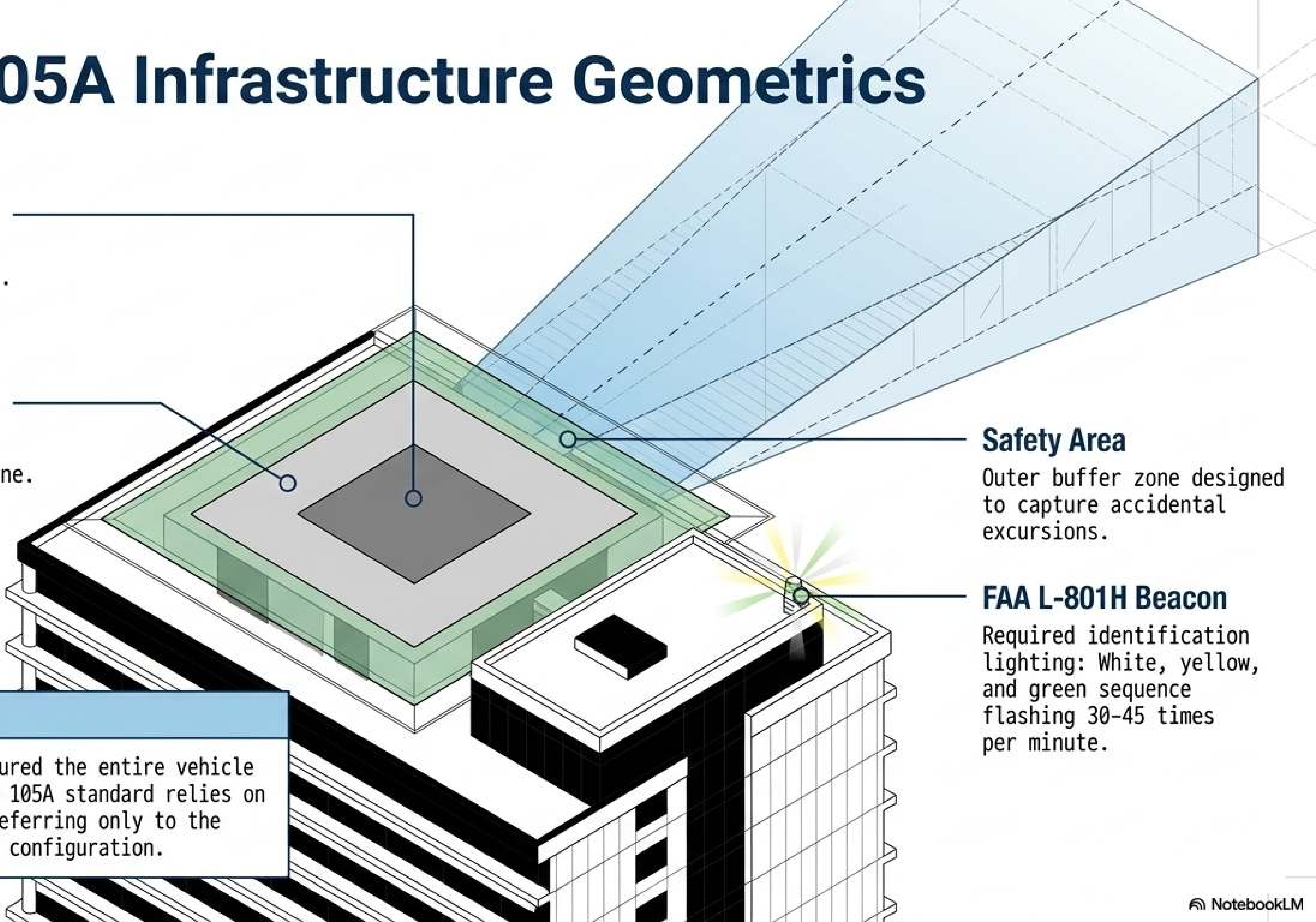
Transitioning from manual flight mechanics to highly automated, fly-by-wire stability control, drastically reducing pilot workload.

DISTRIBUTED ELECTRIC PROPULSION (DEP)

Lift and thrust achieved via 6–8 independently controllable electric motor/rotor pairs, eliminating single points of mechanical failure.

TARGET OPERATIONAL ENVELOPE	
Altitude:	500–3,000 ft AGL
Range:	60–200 miles
Footprint:	40–50 ft controlling dimension (D)
Capacity:	2–6 passengers or cargo equivalent

FAA EB 105A Infrastructure Geometrics



TLOF (Touchdown and Liftoff Area)

Load-bearing center zone. Sized at ≥ 1.0 RD (Rotor Diameter).

FATO (Final Approach and Takeoff Area)

Obstacle-free routing zone. Sized at ≥ 2.0 RD.

Safety Area

Outer buffer zone designed to capture accidental excursions.

FAA L-801H Beacon

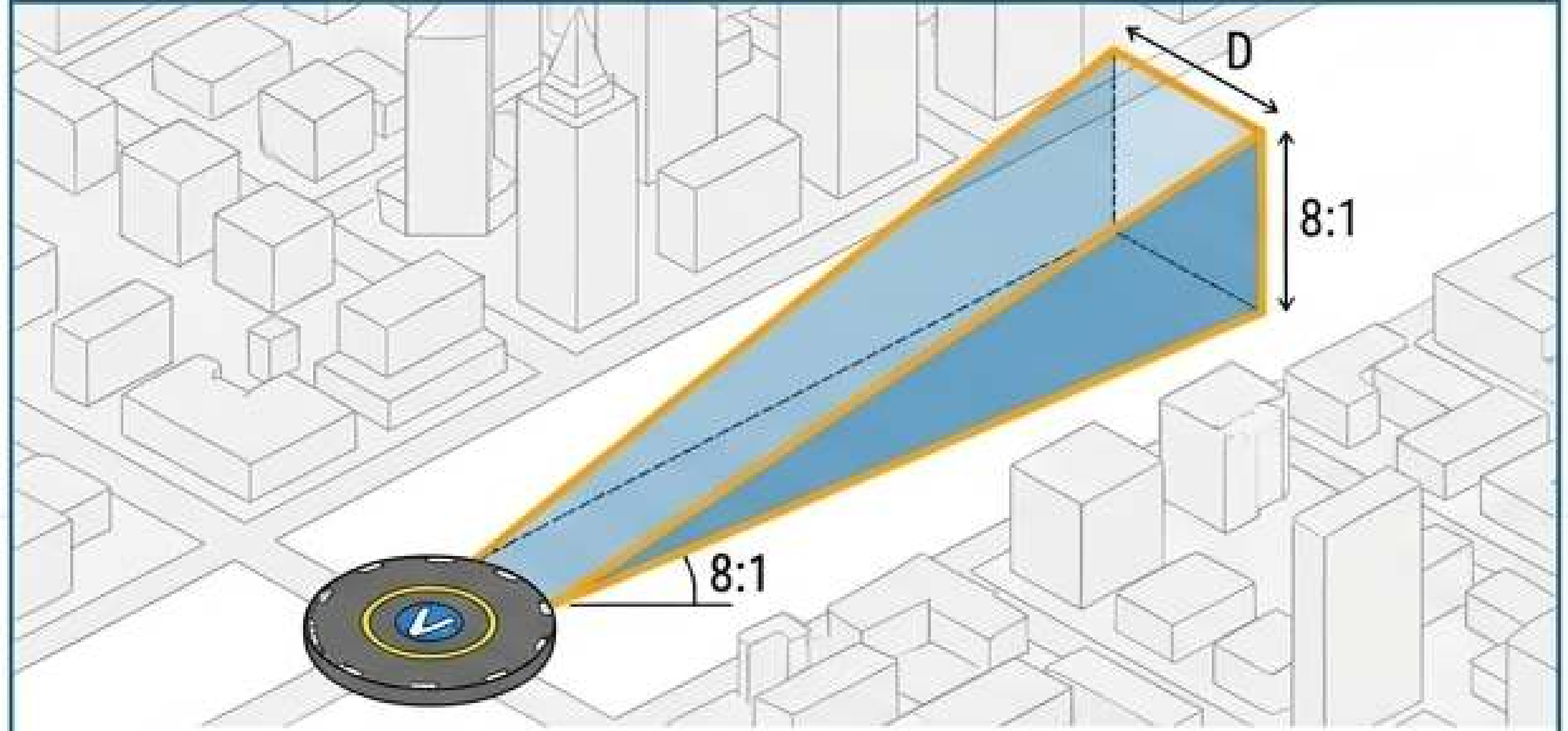
Required identification lighting: White, yellow, and green sequence flashing 30-45 times per minute.

The Geometric Shift

Legacy standard ‘D’ measured the entire vehicle length/width. The new EB 105A standard relies on ‘RD’ (Rotor Diameter), referring only to the propulsion units in VTOL configuration.

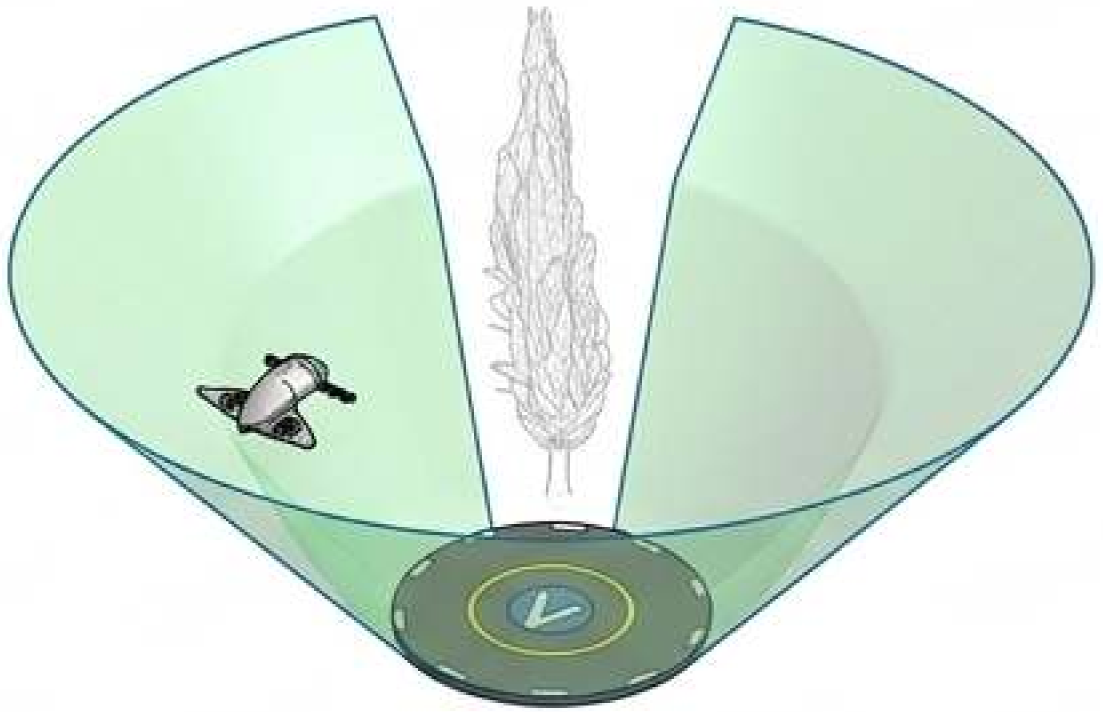
NAVIGATING OBSTACLE CLEARANCES AND APPROACH VOLUMES

FAA TRADITIONAL APPROACH

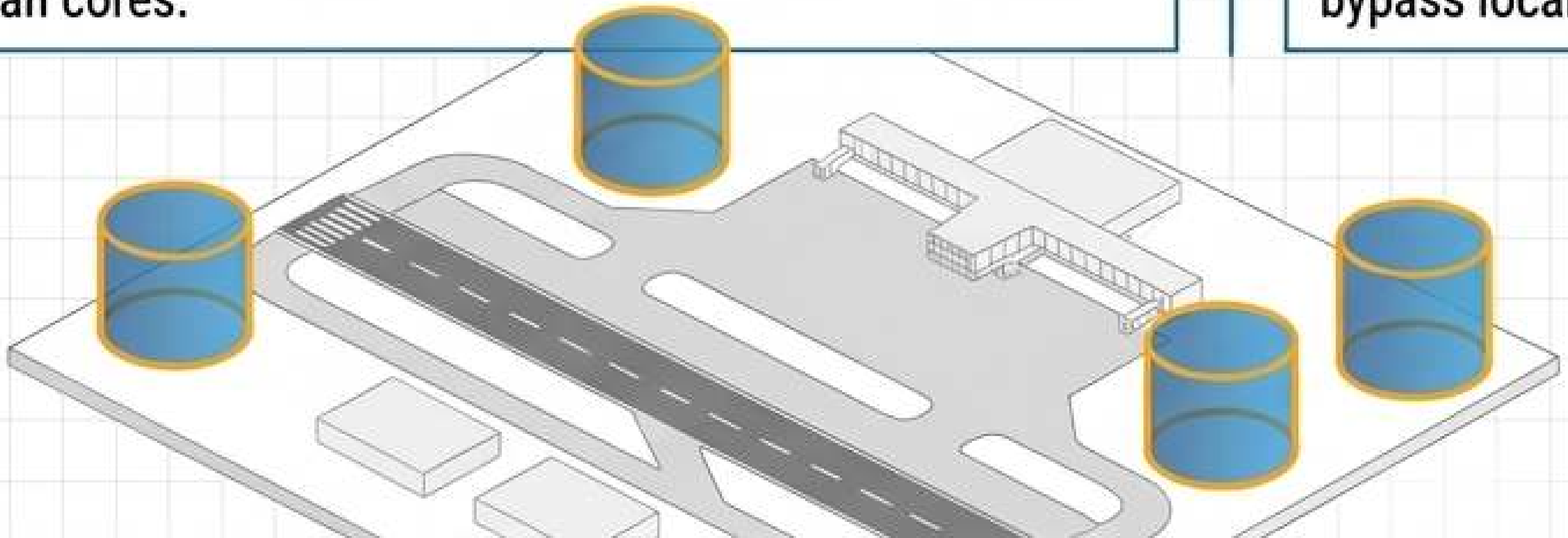


FAA Standard: 8:1 Approach/Departure Slope (8 horizontal units to 1 vertical unit). Requires massive linear clearance, difficult in dense urban cores.

EASA OMNIDIRECTIONAL FUNNEL



EASA Standard (PTS-VPT-DSN): Funnel-shaped omnidirectional obstacle-free volume. Enables steep, flexible approaches that bypass localized noise and structural restrictions.



THE KEYHOLE STRATEGY

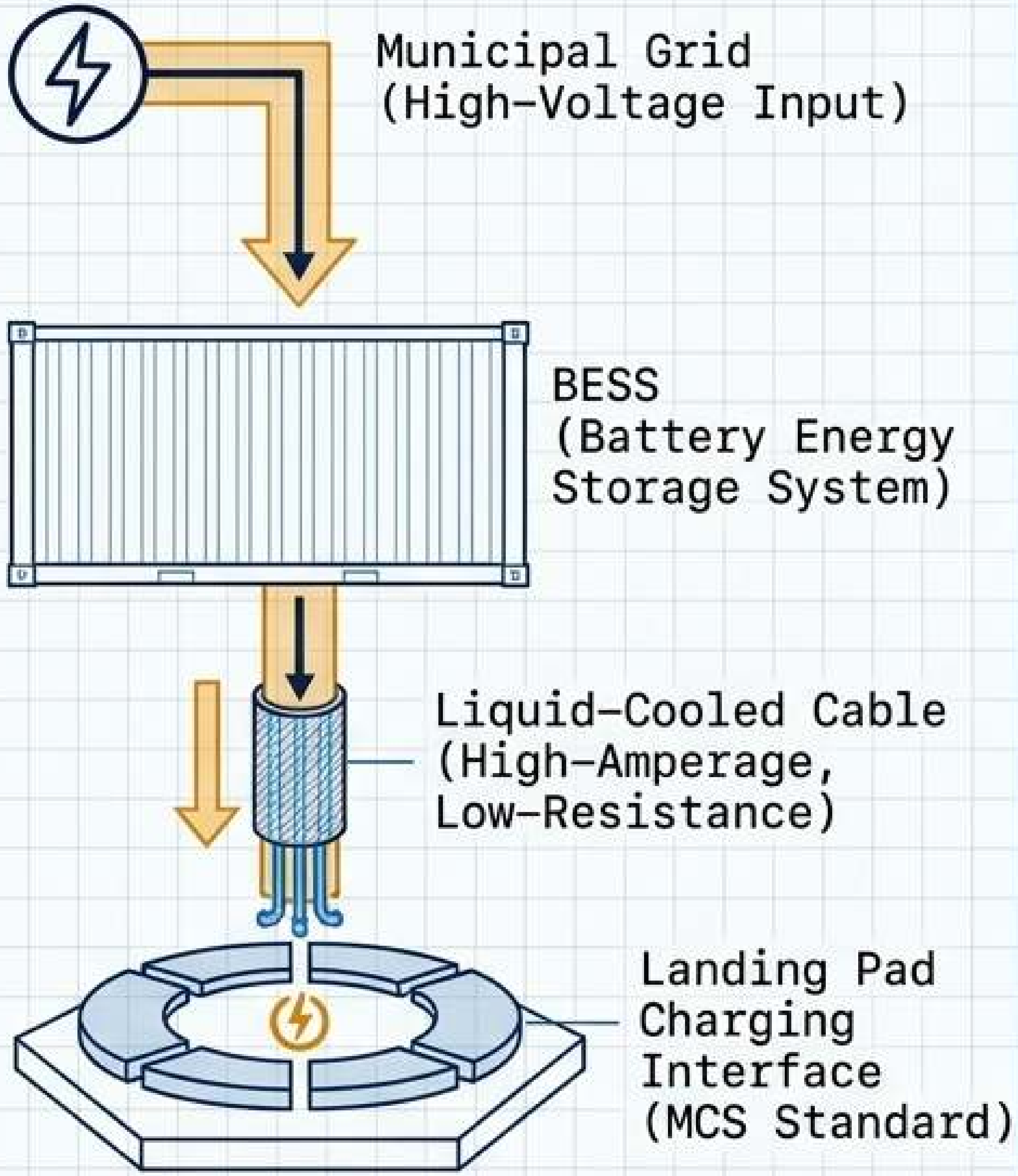
Carving out specific terminal airspace cylinders above vertiports near legacy airports. This omits UAM operations from traditional tower authority, allowing independent, non-interfering vertical traffic flows.

Heliport vs. Vertiport Diagnostic Matrix

Severing the assumption that a vertiport is merely a repainted helipad.

	Legacy Part 139 Heliports	Next-Gen Vertiports
Energy Supply	Jet A liquid fuel storage tanks and traditional refueling trucks.	Megawatt Charging Systems (MCS) & localized Battery Energy Storage Systems (BESS).
Surface Control	Manual Fixed-Base Operator (FBO) authorization and visual marshaling.	Vertiport Automation Systems (VAS) controlling algorithmic slot reservations and FATO scheduling.
Noise Profile	High acoustic footprint, requiring significant zoning buffers.	Lower acoustic footprint enabling deep penetration into population-dense urban centers.
Operational Tempo	Low-volume, ad-hoc, and highly variable scheduling.	High-density, predictive algorithmic turnaround and continuous fleet orchestration.
Safety Protocols	Standard chemical foam Aircraft Rescue and Firefighting (ARFF).	Thermal runaway mitigation, liquid-cooled cabling, and NFPA 855 protocols for lithium-ion fires.

Megawatt Charging Systems and Grid Interface



Megawatt Charging Requirements by OEM

OEM	Power	Current
Lilium	Power: ~1 MW via 20kV downward transformation	Current: 1200A @ 900V
Beta	Power: 350 kW continuous	Current: 350A (Boost t 500A), up to 950 Vdc
Joby	Power: 480V, 3 Phase	Current: 300 Amps

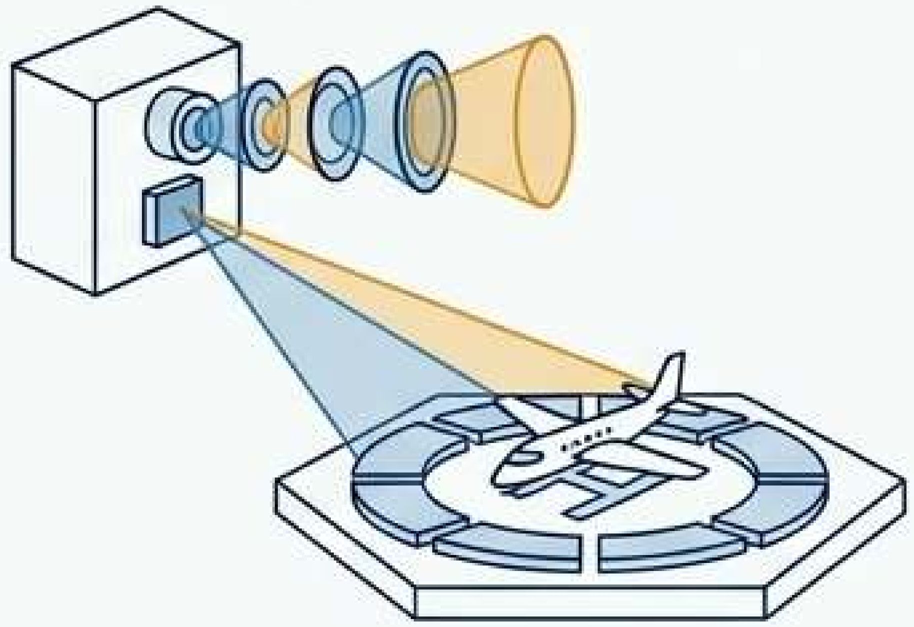
Grid & Thermal Safety Standards

SAE J3271 specification dictates 1.0–2.0 MW per pad.
Thermal management strictly governed by NFPA 855 and UL 9540A fire safety protocols.
Liquid-cooled cables are mandatory to manage heat at high-resistance connection points.

Vertiport Automation Systems (VAS) Architecture

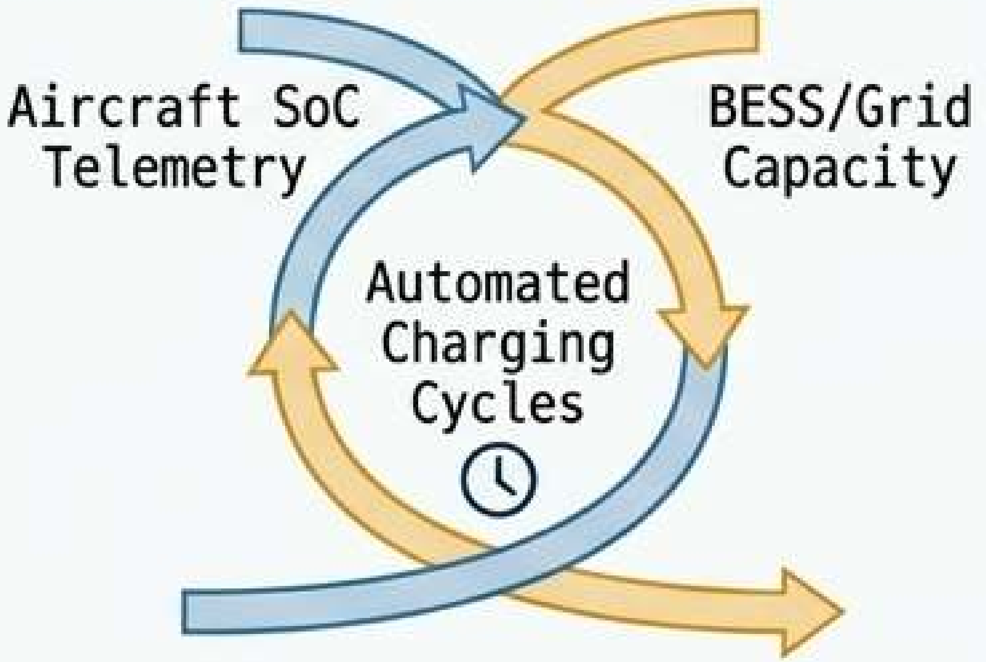
The Digital Operating System for High-Tempo Ground Environments.

Conformance Monitoring



Continuous utilization of **optical and LiDAR sensor arrays** on FATOs. Automates aircraft touchdown detection, verifies **precise parking occupancy**, and triggers instant alerts for **FOD** (Foreign Object Debris).

Ground Resource Management

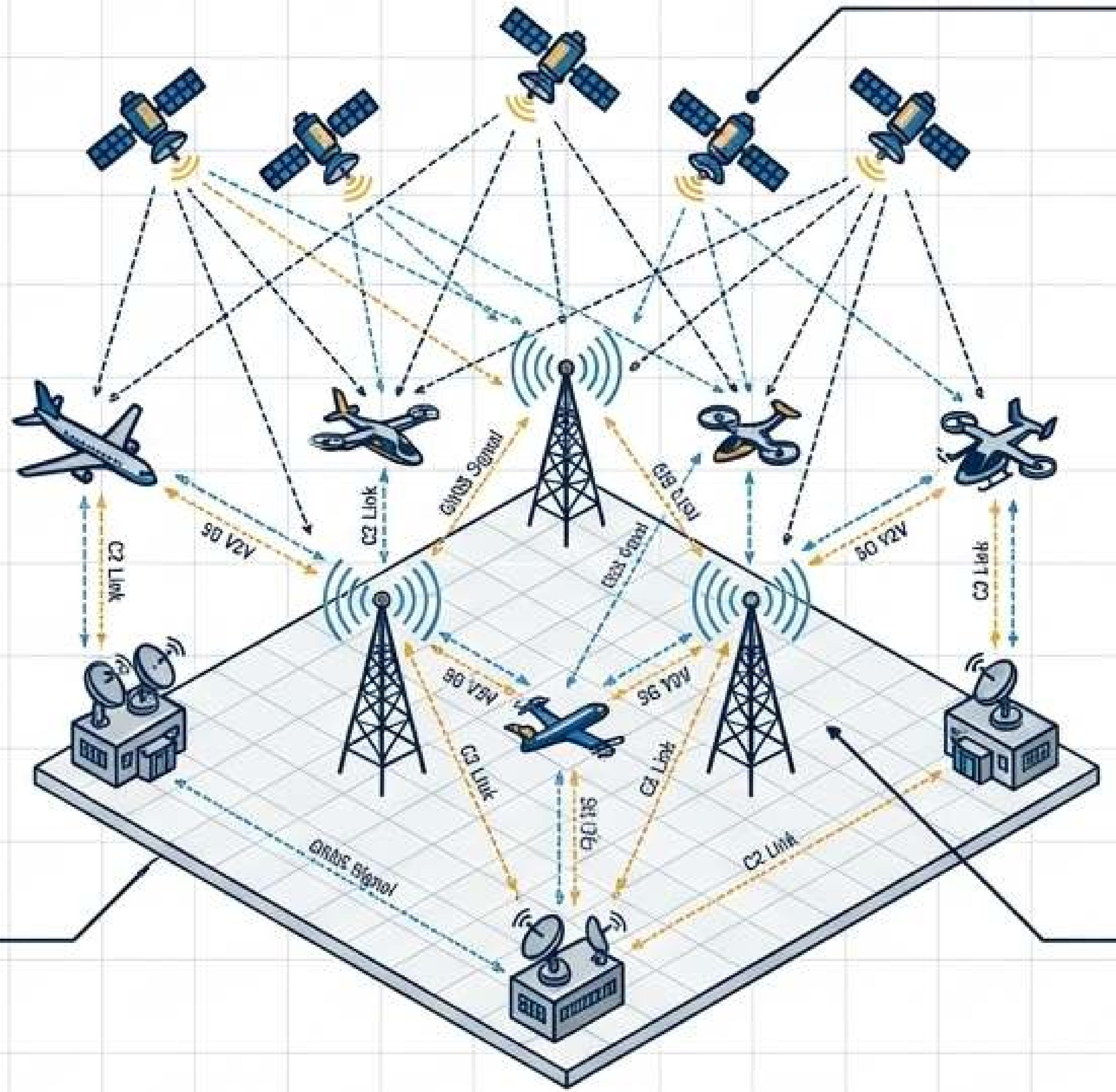


Dynamic matching of energy demands. The VAS orchestrates **automated charging cycles** synchronized to incoming aircraft **State-of-Charge (SoC) telemetry** and local BESS/grid capacity.

Emergency Management

Real-time thermal oversight. Actively monitors battery temperatures during megawatt charging. Configured to automatically trigger **localized suppression systems** in the event of **lithium-ion thermal runaway**.

Integrated CNS and Spectrum Telemetry



Navigation

Transition to dual-frequency GNSS and augmented integrity systems.

Required for precision UAM RNP route-following within drastically reduced separation minima in urban corridors.

The Integration Imperative

Legacy aviation separated Communications, Navigation, and Surveillance.

High-density AAM forces the convergence of all three into a singular, integrated digital airspace gridding structure.

Communications

Leveraging 5G V2V (Vehicle-to-Vehicle) datalinks.

Enables continuous state reporting, extremely low latency C2 links, and onboard Detect and Avoid (DAA) processing.

Algorithmic Demand-Capacity Balancing

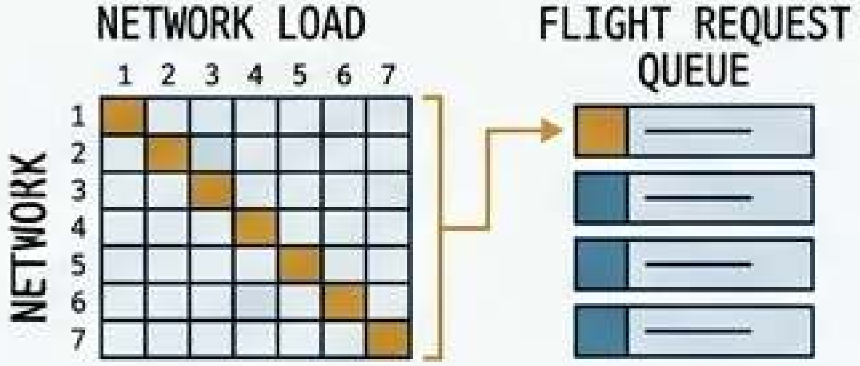
The Threat of Risk Singularities

Relying on legacy “first-come, first-served” (FCFS) ATC models in high-density UAM creates risk singularities—bottlenecks of battery-constrained eVTOLs stacked in hovering holding patterns.

Stage 1: Pre-Flight Planning

Strategic Conflict Management Services (SCMS) evaluate network load. Vertiport Managers (VMs) process flight requests against known pad availability.

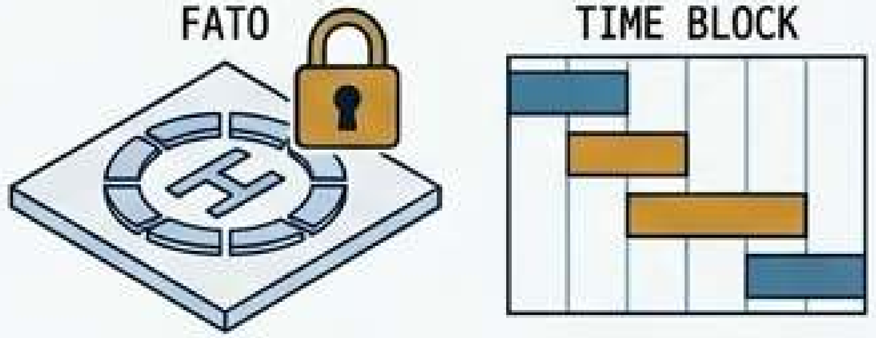
PAD_AVAILABILITY_CHECK



Stage 2: Resource Assurance & FATO Scheduling

Absolute resource lock. Departure is strictly prohibited until the destination FATO reservation and exact turnaround window are algorithmically guaranteed.

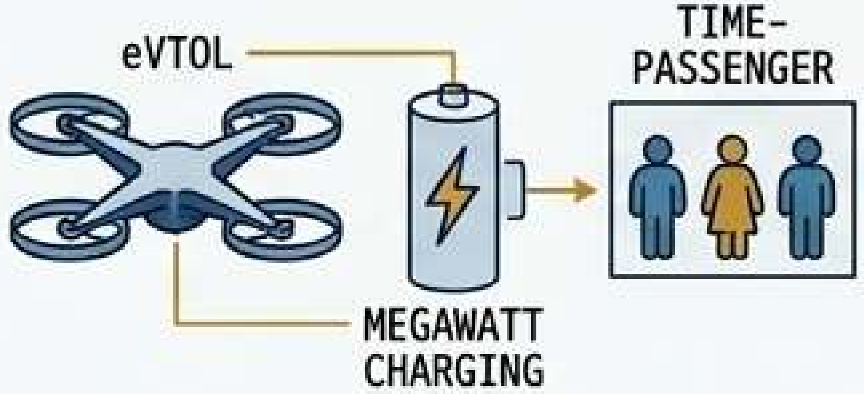
TURNAROUND_WINDOW_GUARANTEED
DEPARTURE_PROHIBITION: ACTIVE



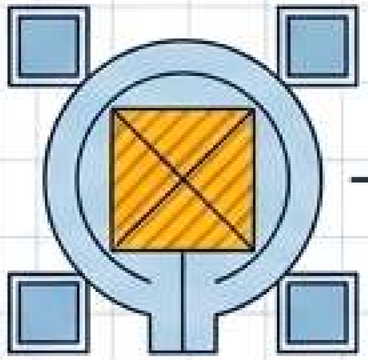
Stage 3: Execution & Turnaround

Aircraft arrives exactly on time. Megawatt charging and passenger swap occur within a rigidly controlled, predictive time block before immediate egress.

IMMEDIATE_EGRESS



Urban Operational Topology

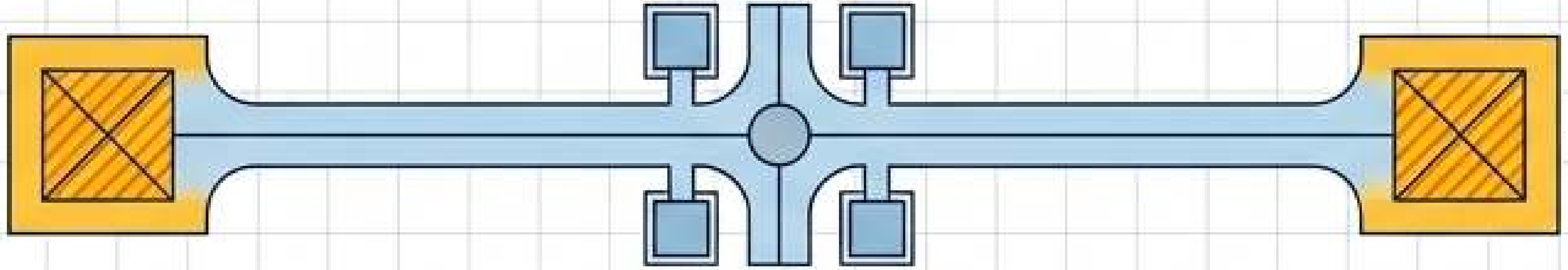


MINIMUM FOOTPRINT VERTIPORT LAYOUT

Type: VERTISTOP

Scale: 1 landing site. Minimal physical infrastructure.

Role: Dedicated exclusively to immediate passenger/cargo boarding and discharge. No permanent charging or maintenance capabilities.

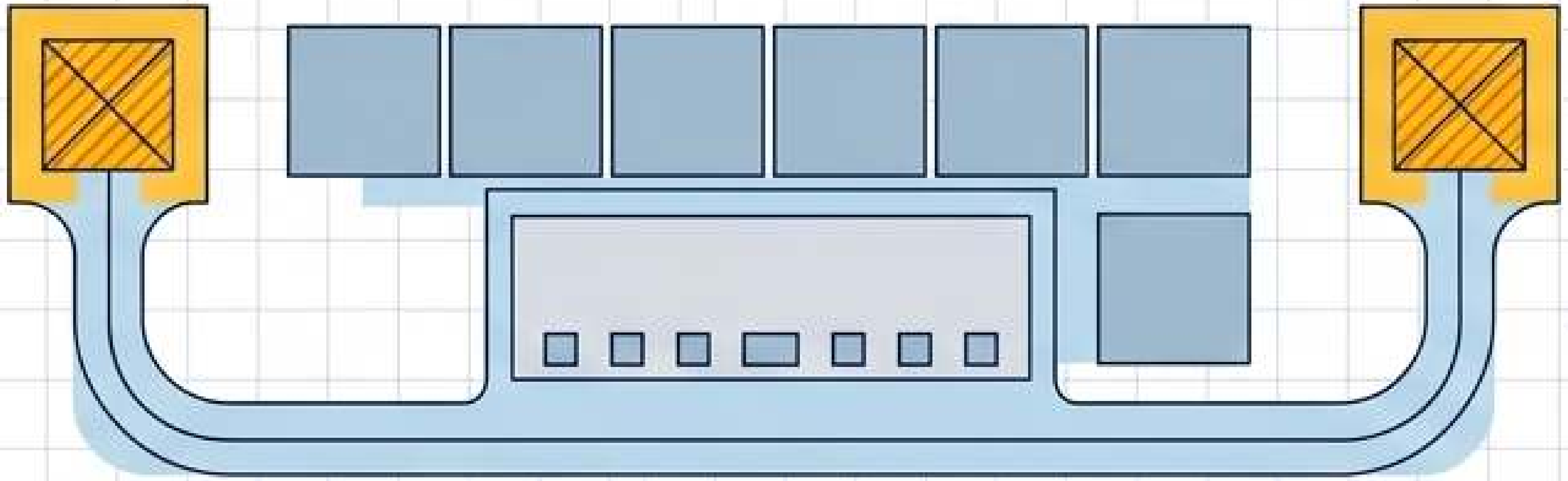


LINEAR/DRIVE-THROUGH VERTIPORT LAYOUT

Type: VERTIPORT

Scale: 2+ landing sites with adjacent gates and taxiways.

Role: Equipped with Megawatt Charging Systems (MCS), automated passenger terminal infrastructure, and dedicated Vertiport Manager (VM) oversight.



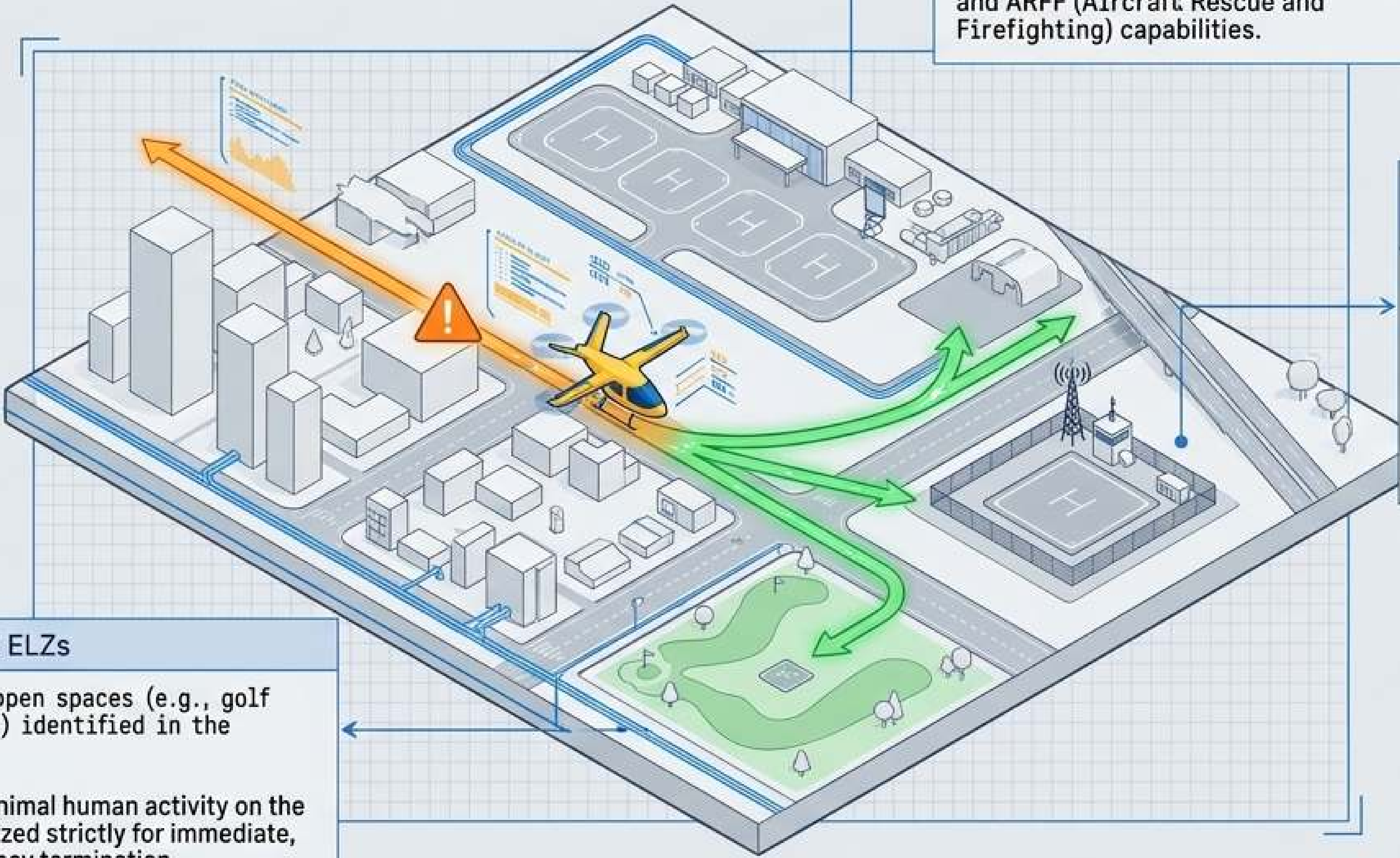
AIRPORT-LIKE VERTIPORT LAYOUT

Type: VERTIHUB

Scale: 2+ landing sites, extensive parking, and large hangars.

Role: Serves as the fleet base of operations. Houses NOCC (Network Operations Control Center) and heavy MRO (Maintenance, Repair, and Overhaul) capabilities.

Off-Nominal Procedures and Contingency Routing



DIVERSION VERTIPORTS

Standard vertiports within the network holding algorithmic excess capacity.
Preferred for off-nominal landings as they possess full passenger services and ARFF (Aircraft Rescue and Firefighting) capabilities.

SECURE EMERGENCY LANDING ZONES (SELZ)

Purpose-built secure sites placed strictly along UAM RNP routes where full vertiports are geographically unavailable.

Equipped with fundamental PNT (Positioning, Navigation, and Timing) and C2 datalink coverage.

NON-SECURE ELZs

Pre-surveyed open spaces (e.g., golf courses, parks) identified in the digital twin.

Selected for minimal human activity on the ground and utilized strictly for immediate, critical emergency termination.

Transitioning to the Autonomous Horizon

UAM Maturity Levels and the Evolution of the Pilot

1 Current State: SVO

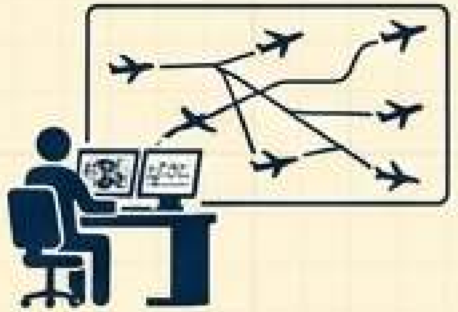
Simplified Vehicle Operations. An onboard human pilot physically manages flight computers, though direct manual linkage to rotors is replaced by fly-by-wire stability.

2 Mid-Term: HOTL & HOVTL

Human-On-The-Loop: Automation handles standard flight; human intervenes only for contingency navigation.

Human-Over-The-Loop: Automation completely manages inner-loop stability and control. The vehicle is supervised, not flown.

3 Future State: MVS

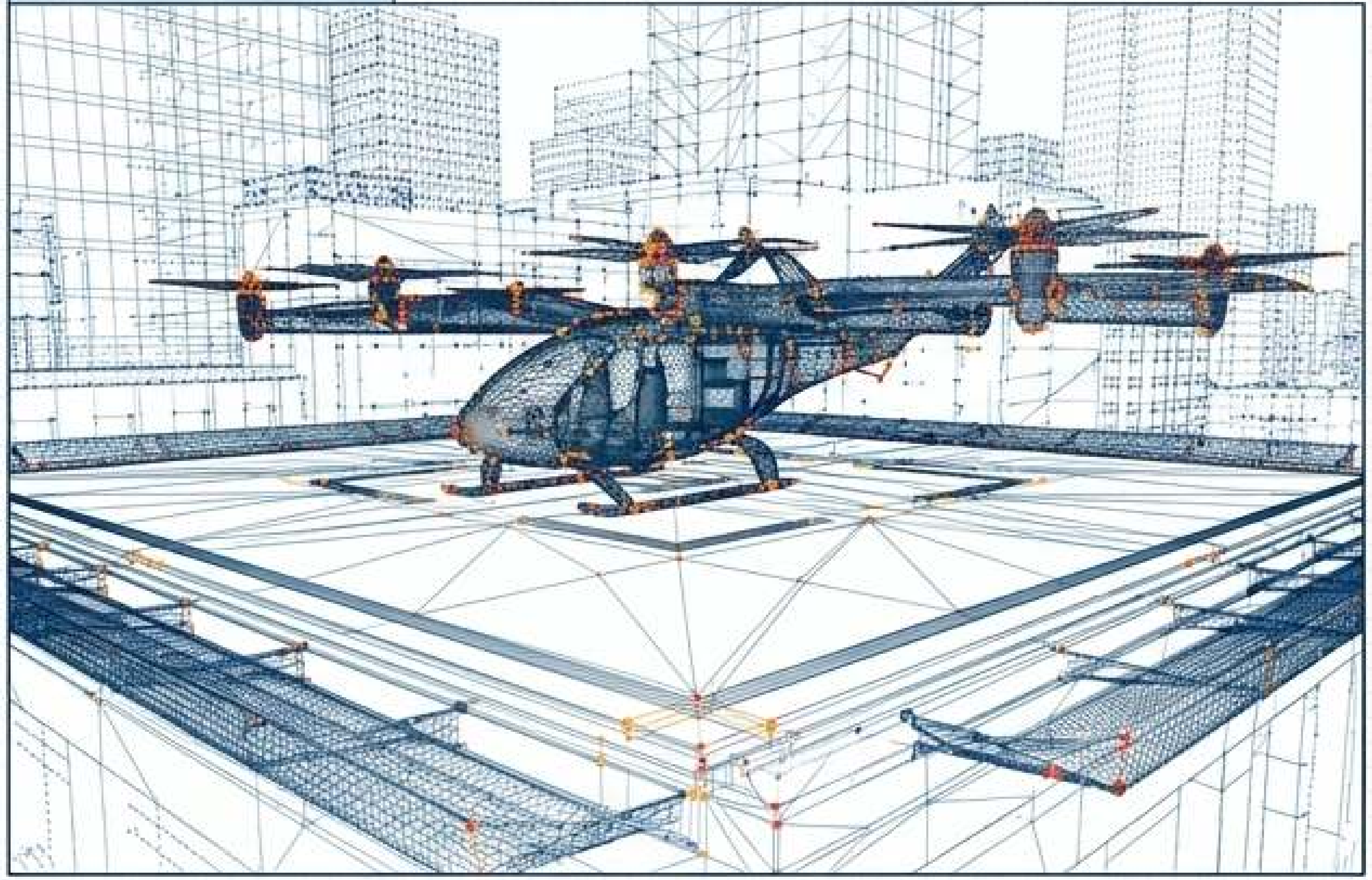


Multi-Vehicle Supervisor. A single ground-based human securely overseeing 3 to many uncrewed aircraft. Vehicles automatically execute complex complex flight plans along defined UAM RNP routes, shifting human interaction from pilot to algorithmic orchestrator.

Digital Twin Simulation Frameworks

PHYSICAL ASSET

DIGITAL TWIN SIMULATION



STATUS: NOMINAL // ALT: 000.0m // LAT: 37.422° N // LON: 122.084° W // BAT: 98.5% // POWER FLOW: +0.2kW // WIND: 3.5 m/s NE // SYSTEM HEALTH: OPTIMAL // SIMULATION TIME: 14:05:22 UTC // DATA RATE: 10 Gbps // COMPUTE LOAD: 45% // ERROR PROBABILITY: <0.001%

Engine Integration

High-fidelity artificial systems are built utilizing Unreal Engine, AirSim, and Cesium to create 1:1 operational models of urban vertiports before physical concrete is poured.

Dynamic Risk Assessment

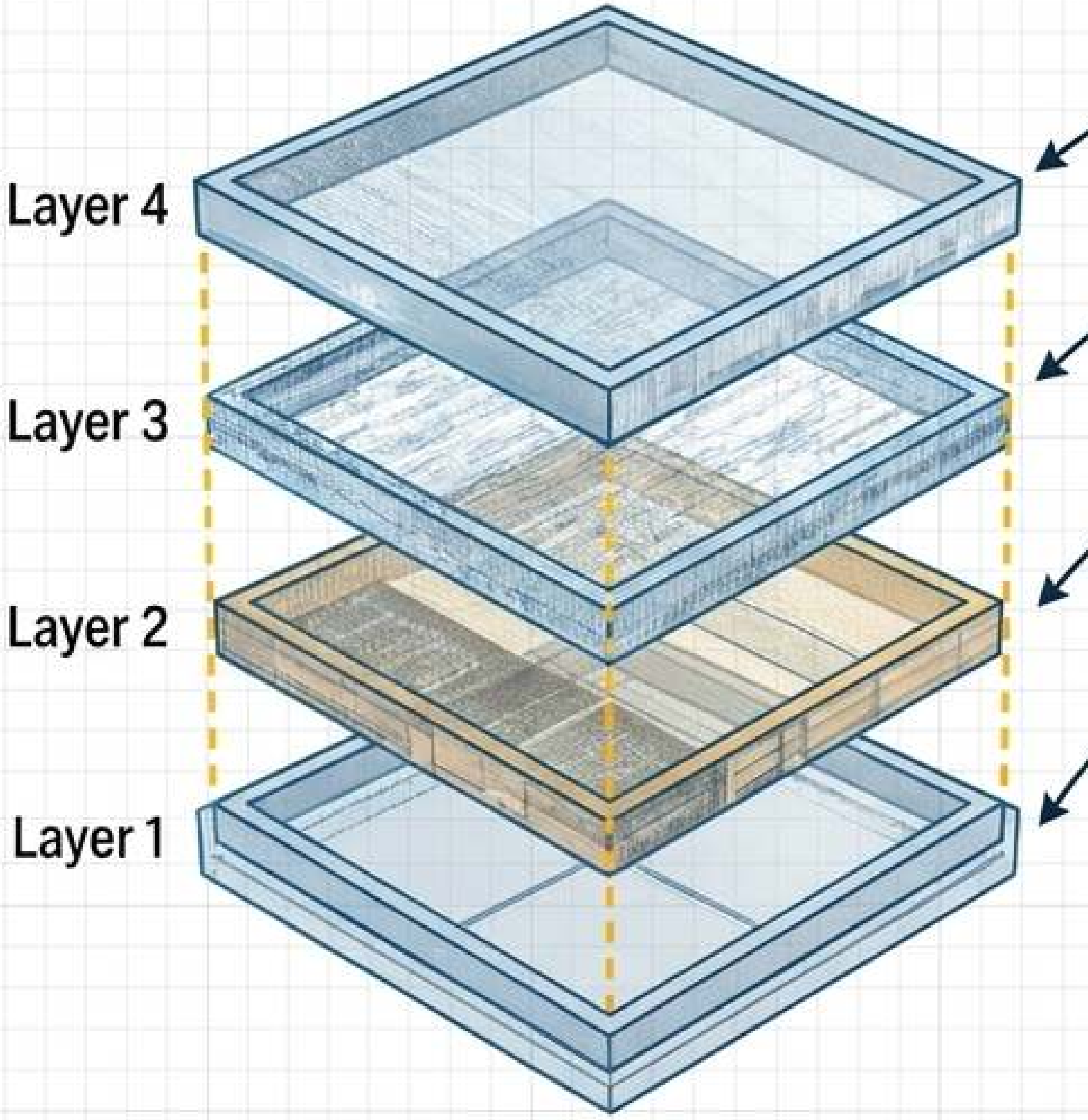
Simulates nominal and severely disrupted conditions (adverse micro-weather, engine failures). Allows safe, zero-risk evaluation of FATO capacity limits and turnaround delays.

Algorithmic Airspace

Ingests live environmental data to project dynamic geofences over the virtual city, enabling the testing of orderly route re-planning algorithms.

The Fully Integrated Mobility Ecosystem

UAM is a Cyber-Physical System, Not Just an Aircraft.



Application Layer
High-Tempo Logistics & Demand-Capacity Balancing (SCMS).

Data & CNS Layer
Vertiport Automation Systems (VAS) & 5G Telemetry Grid.

Power Layer
Megawatt Charging Systems (MCS) & Localized BESS Interfaces.

Physical Layer
FAA EB 105A Geometric Pad Design & eVTOL Hardware.

Synthesis Insight:

Achieving high-volume Urban Air Mobility relies entirely on the precise algorithmic synchronization of ground power and digital traffic management, rather than the isolated aerodynamic capabilities of the eVTOL platform itself.

Critical Regulatory and Technical Specifications

Design Standards

- **FAA EB 105A**
  Interim Vertiport Design guidelines for VFR conditions.

- **EASA PTS-VPT-DSN**
  Prototype Technical Specifications for VFR Vertiports, defining omnidirectional approach funnels.

Power & Data Standards

- **SAE J3271**
  Megawatt Charging System (MCS) standard for heavy electric aviation.

- **NFPA 855 / UL 9540A**
  Strict fire safety testing protocols for Battery Energy Storage Systems (BESS).

- **RTCA DO-362A / DO-373**
  Command and Control (C2) Datalink and GNSS Navigation integration parameters.

Core Geometrics

- **TLOF Sizing:**
  Load-bearing center area must be ≥ 1.0 RD.

- **FATO Sizing:**
  Obstacle-free approach zone must be ≥ 2.0 RD.

- **Dimensional Shift:**
  Moving away from traditional “D” length to “Rotor Diameter (RD)” to define bounding boxes.